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REVIEW

Leadless Pacemakers in the Setting of Surgical and Transcatheter Tricuspid Valve Procedures

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ABSTRACT: Transvenous pacing is increasingly recognized as problematic in patients with prior or concomitant tricuspid valve intervention, owing to risks of leaflet interference, prosthetic dysfunction, and progression of tricuspid regurgitation. Leadless pacemakers offer a valve-sparing alternative; however, their safety and performance in structurally altered right heart anatomy remain incompletely defined. We conducted a systematic review to evaluate procedural feasibility, electrical performance, device-valve interaction, and clinical outcomes of leadless pacemaker implantation in patients undergoing surgical or transcatheter tricuspid valve interventions. Thirty-four studies comprising 272 patients were included, encompassing surgical repair, bioprosthetic replacement, valve-in-valve procedures, transcatheter edge-to-edge repair, and transcatheter tricuspid valve replacement. Leadless pacemakers were implanted via transfemoral, transjugular, or direct surgical approaches, achieving a procedural success rate of 99.3%. Electrical performance was consistently favorable, with stable capture thresholds, sensing amplitudes, and impedance during follow-up. Device-valve interaction was infrequent and generally manageable. Leadless pacemaker–related complications were rare (1.1%), with no device-related mortality. Within the included studies of patients undergoing surgical or transcatheter tricuspid valve interventions, no study demonstrated worsening tricuspid regurgitation attributable to the leadless pacemaker or its delivery system. Across a broad spectrum of complex tricuspid anatomies, leadless pacemakers demonstrated excellent feasibility, durable electrical performance, and a low complication profile, supporting their role as a valve-sparing pacing strategy in this population. These findings support leadless pacing in patients with prior tricuspid intervention, although prospective comparative data are required to define its role relative to alternative pacing modalities.

Key Words: electric impedance ■ follow-up studies ■ humans ■ prospective studies ■ tricuspid valve

Tricuspid regurgitation (TR) is increasingly recognized as a major cause of morbidity and mortality, yet remains underdiagnosed and undertreated compared with left-sided valve disease. Its prevalence continues to rise in parallel with the expanding use of cardiac implantable electronic devices, especially transvenous pacemakers and defibrillators, which are now identified as a leading cause of secondary (lead-induced) TR.[1,2] Mechanical interference between right ventricular leads and the tricuspid valve apparatus—including leaflet impingement, entanglement within chordae, papillary muscle tethering, annular distortion, and leaflet perforation—can precipitate or worsen TR, accelerate right-sided remodeling, and contribute to heart failure hospitalizations and reduced survival.[3,4] Both the European Heart Rhythm Association and American College of Cardiology now explicitly acknowledge lead-related TR as a distinct disease process requiring dedicated evaluation and tailored pacing strategies.[2,5]

As tricuspid valve interventions—including surgical repair, surgical bioprosthetic replacement, valve-in-valve procedures, transcatheter edge-to-edge repair (TEER), and transcatheter tricuspid valve replacement (TTVR)—become increasingly common, the need for permanent pacing is rising. Conduction disturbances, including high-grade atrioventricular block and sinus node dysfunction, occur frequently because annuloplasty rings, prosthetic valves, and large-frame TTVR systems lie in close proximity to the atrioventricular node and His bundle.[4,6]

However, conventional transvenous pacing in these patients is associated with unique risks. Leads crossing repaired or prosthetic tricuspid valves may cause:

- leaflet or commissural injury, impairing repair durability[1,4]

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Nonstandard Abbreviations and Acronyms

LP	leadless pacemaker
TEER	transcatheter edge-to-edge repair
TR	tricuspid regurgitation
TTVR	transcatheter tricuspid valve replacement

- restriction of bioprosthetic leaflet mobility, promoting early structural valve degeneration[3,5]
- jailing between TEER clips or TTVR frames, complicating extraction and predisposing to lead dysfunction[2,5]
- entrapment, making extraction hazardous or impossible, particularly in pacemaker-dependent patients[2,5]
- prosthetic valve or TTVR frame damage during lead advancement or extraction attempts[4,6]

Transvenous leads also substantially increase the risk of device-related endocarditis—especially catastrophic in patients with prosthetic or transcatheter tricuspid devices—and extraction in this setting carries prohibitive morbidity.[7–9] Accordingly, guideline documents emphasize that postintervention tricuspid anatomy constitutes a high-risk substrate in which traditional transvenous leads may compromise valve integrity, procedural success, and long-term outcomes.[2,5,10]

Epicardial and coronary sinus pacing systems avoid crossing the tricuspid valve but are limited by higher chronic thresholds, reduced sensing reliability, lead instability, and higher long-term failure rates.[11,12] Surgical epicardial leads require thoracotomy or sternotomy, making them undesirable for frail patients, while coronary sinus pacing is constrained by venous anatomy and often provides suboptimal ventricular capture.[11,12]

Leadless pacemakers (LPs) offer a fundamentally different, valve-sparing strategy by eliminating the need for transvalvular leads. Contemporary LP systems demonstrate excellent safety and stable long-term performance in the general population.[13] Yet, their performance in patients with prior or concomitant tricuspid interventions is less well defined. Annuloplasty rings, bioprosthetic valves, TEER clips, and large-frame TTVR devices may impede delivery catheter navigation or influence device positioning; conversely, LPs may reduce TR progression by avoiding direct leaflet interaction.[14] Current consensus statements highlight the theoretical advantages of LPs but emphasize the major gap in systematic evidence specific to this population.[15,16]

Given the rapidly expanding population undergoing tricuspid repair or replacement—and increasing recognition that conventional pacing strategies may compromise valve function and long-term outcomes—there is a critical need for a comprehensive synthesis of the available evidence. This systematic review aims to evaluate all published data on leadless pacemaker implantation in the setting of surgical or transcatheter tricuspid valve procedures, characterizing procedural success, valve interaction, electrical performance, and clinical outcomes, and defining the emerging role of LPs in contemporary tricuspid valve management. To our knowledge, no prior systematic review has specifically focused on leadless pacemaker implantation in the context of surgical and transcatheter tricuspid valve interventions.

METHODS

Search Strategy

This systematic review was performed according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines.[17,18] A comprehensive search was conducted across PubMed/MEDLINE, Embase, Scopus, Web of Science, CINAHL, and the Cochrane Library from inception to March 12, 2026.

The PubMed strategy was:

("Micra" OR "Aveir" OR "Nanostim" OR "leadless pacemaker") AND

("tricuspid valve repair" OR "tricuspid valve replacement" OR "bioprosthetic tricuspid" OR "valve-in-valve" OR "TTVR" OR "TriClip" OR "Pascal" OR "edge-to-edge" OR "tricuspid intervention").

Search strategies were adapted for all databases using relevant controlled vocabulary.

No restrictions were placed on language, publication year, or geographic region.

Reference lists of all eligible full texts and relevant narrative reviews were manually screened to identify additional studies.

Eligibility Criteria

Studies were eligible if they included human subjects and reported:

1. Implantation of an LP of any generation (Micra VR/AV, Aveir VR, Nanostim).
2. A prior, concomitant, or subsequent tricuspid valve intervention, including:
   - Surgical tricuspid valve repair
   - Surgical tricuspid valve replacement
   - Transcatheter valve-in-valve procedures
   - TEER (TriClip or Pascal)
   - Orthotopic TTVR
3. Extractable data on demographics, procedural details, device performance, valve interaction, or clinical outcomes.

Eligible study designs included:

- Randomized or nonrandomized cohort studies
- Prospective or retrospective case series
- Case reports

Exclusion criteria:

- Reviews, editorials, and expert commentary
- Animal or bench-top studies
- Conference abstracts lacking analyzable data
- Studies not specifying the relationship between LP implantation and the tricuspid valve intervention
- Reports describing only conventional transvenous pacemakers

Study Selection Process

All retrieved records were imported into EndNote, and duplicates were removed.

The remaining studies underwent title and abstract screening by 2 independent reviewers.

Studies were excluded for:

- No leadless pacemaker
- No tricuspid intervention
- Transvenous pacing only
- Insufficient procedural detail

Disagreements between reviewers were resolved by consensus. The study selection process is summarized in the Figure.

Data Extraction

Two reviewers independently extracted data using a structured template, capturing:

- Study characteristics (design, country, year)
- Patient demographics and comorbidities
- Tricuspid intervention type (surgical/TEER/TTVR/valve-in-valve)
- Leadless pacemaker type, approach route, and procedural success
- Electrical performance (acute and follow-up)
- Valve-device interaction
- Complications (procedure-related and device-related)
- Follow-up outcomes

No authors were contacted for missing data.

Data were extracted as reported; no imputation was performed.

Quality Assessment

The methodological quality of included studies was appraised using established, design-appropriate frameworks. For case reports and case series, a modified Murad et al[19] tool was applied, assessing clarity of the case definition, adequacy of exposure and outcome ascertainment, completeness of procedural description, reporting of key electrical performance parameters (capture threshold, sensing amplitude, impedance), consideration of alternative explanations, and adequacy of follow-up. For observational cohort studies, quality was assessed using a modified Newcastle-Ottawa Scale adapted to single-arm device studies, evaluating cohort representativeness, exposure ascertainment, outcome assessment, follow-up adequacy, and completeness of procedural and electrical reporting. The domain-level assessment for each included study is summarized in Table 1.

Data Synthesis and Analysis

Due to the heterogeneity of study designs, populations, and reporting formats, a meta-analysis was not appropriate. Thus, a narrative and quantitative synthesis was performed.

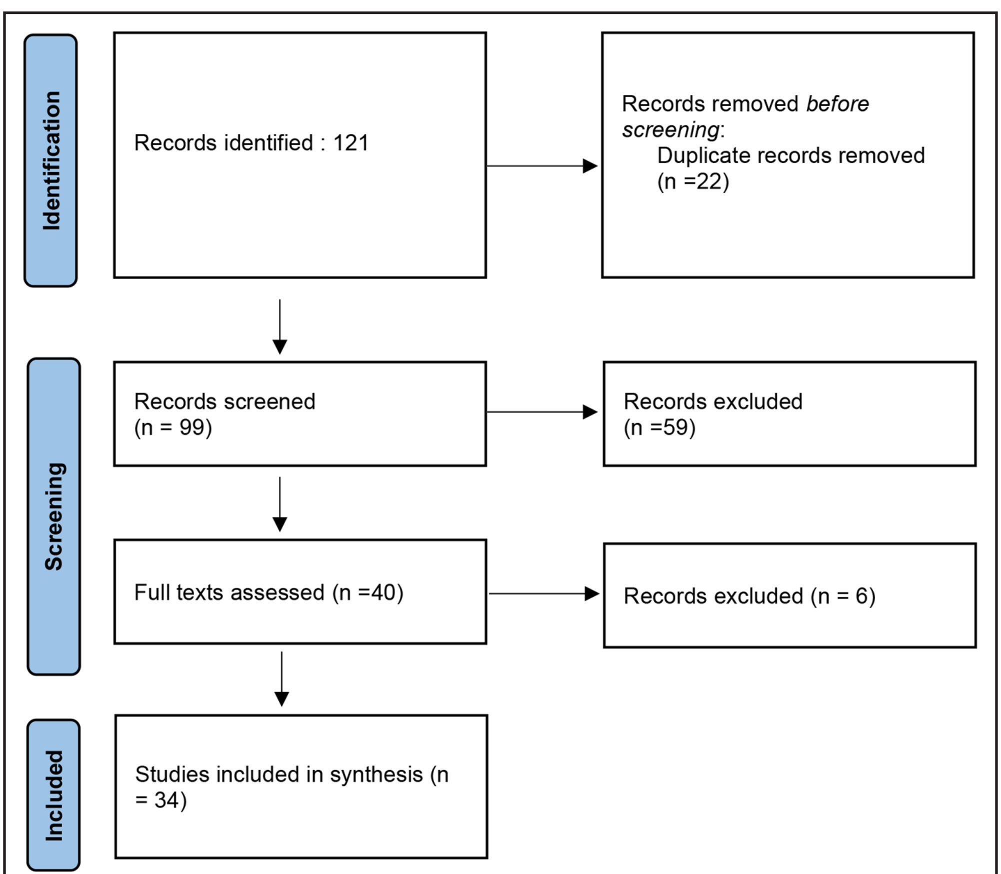

Figure. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flowchart of study selection.
Flowchart showing identification, screening, eligibility assessment, and final inclusion of studies evaluating leadless pacemaker implantation in patients with surgical or transcatheter tricuspid valve interventions.

Table 1. Methodological Quality Assessment of Included Studies

Study (first author, year)	Study design	Procedural detail reported	Electrical parameters reported	Outcome ascertainment	Adequate follow-up	Overall quality
Terricabras, 2021[20]	Case report	Yes	Partial	Yes	Yes	High
Garweg, 2020[21]	Case report	Yes	Partial	Yes	Yes	High
Hale, 2022[22]	Case report	Yes	Partial	Yes	Yes	High
Mohamed, 2025[23]	Case report	Yes	Partial	Yes	Yes	High
Afzal, 2019[24]	Multicentre cohort	Yes	Partial	Yes	Yes	Good–high
Doyle, 2025[25]	Case report	Yes	Partial	Yes	Yes	High
Mohammadi, 2025[26]	Case report	Yes	Partial	Yes	Yes	High
Adukauskaite, 2020[27]	Case report	Yes	Partial	Yes	Yes	High
Fink, 2024[28]	Case report	Yes	Yes	Yes	Partial	High
Morani, 2021[29]	Case report	Yes	Yes	Yes	Partial	High
Ozdemir, 2025[30]	Case report	Yes	Partial	Yes	Yes	High
Augustin, 2025[31]	Case report	Yes	Yes	Yes	Yes	High
Rangavajla, 2025[32]	Cohort	Yes	Partial	Yes	Yes	Good–high
Goto, 2024[33]	Case report	Yes	Yes	Yes	Yes	High
Oates, 2024[34]	Retrospective cohort	Yes	Yes	Yes	Yes	High
Patrascu, 2025[35]	Case report	Yes	Partial	Yes	Yes	High
Shivamurthy, 2022[36]	Retrospective cohort	Yes	Yes	Yes	Yes	High
Arunthamakun, 2024[37]	Case series	Yes	Yes	Yes	Partial	High
Aders & Strobel, 2024[38]	Case report	Yes	Yes	Yes	Yes	High
Demirtakan, 2025[39]	Case report	Yes	Yes	Yes	Partial	High
Vuppuluri, 2025[40]	Case report	Yes	Partial	Yes	Yes	High
Poggio, 2024[41]	Case report	Yes	Yes	Yes	Partial	High
Zou, 2025[42]	Case report	Yes	Yes	Yes	Yes	High
Shen, 2025[43]	Case series	Yes	Partial	Yes	Partial	Moderate–high
Dreyfus, 2023[44]	Case report	Yes	No	Yes	Yes	High
Tang, 2017[45]	Case report	Yes	Partial	Yes	Yes	High
Braunstein, 2025[46]	Retrospective cohort	Yes	Partial	Yes	Yes	High
Tat, 2025[47]	Retrospective cohort	Yes	Yes	Yes	Yes	High
Gul, 2024[48]	Multicentre cohort	Yes	Yes	Yes	Yes	High
Theis, 2020[49]	Retrospective cohort	Yes	Yes	Yes	Yes	High
Fortuna, 2026[50]	Case report	Yes	Yes	Yes	Yes	High
Friedrich, 2025[51]	Retrospective single-center cohort	Yes	Yes	Yes	Yes	High
Segreti, 2025[52]	Case report	Yes	Partial	Yes	Yes	Good–high
Mekary, 2026[53]	Retrospective cohort study	Yes	Yes	Yes	Yes	High

Methodological quality was evaluated using design-specific tools. Case reports and case series were assessed using a modified Murad et al[19] framework, including clarity of case definition, adequacy of exposure and outcome ascertainment, completeness of procedural and device-related reporting, and adequacy of follow-up. Observational cohort studies were assessed using a modified Newcastle-Ottawa Scale adapted for single-arm device studies, evaluating cohort representativeness, exposure ascertainment, outcome assessment, and follow-up completeness. Partial indicates incomplete or nonsystematic reporting of the specified domain.

When raw denominators were available:

- Categorical data were summarized using counts and percentages.
- Continuous variables were reported using means or medians, consistent with the original publications.

Where studies provided overlapping or partially incomplete data, conservative aggregation strategies were used to avoid duplication.

RESULTS

A total of 121 records were identified through database searching. After removal of 22 duplicates, 99 studies underwent title and abstract screening, of which 59 were excluded for irrelevance. Forty full-text articles were assessed for eligibility, and 34 studies met all inclusion criteria and were included in the final synthesis.[20–53] These comprised:

- 20 single-patient case reports,
- 2 small case series (2 patients each), and
- 12 observational cohort studies enrolling between 12 and 100 patients.

Study and Patient Characteristics

Thirty-four studies reporting 272 patients who underwent leadless pacemaker implantation in the context of tricuspid valve intervention are summarized in Table 2. Publications spanned 2017 to 2026 and originated from 15 countries, reflecting broad international experience.

Based on aggregated study-level data, 147 patients (54.0%) were female. Age ranged from 21 to 90 years, with most patients in the seventh or eighth decade of life. Atrial fibrillation was frequent and could be quantified in most reports, occurring in 196 of 272 (72.1%) patients. Other comorbidities—such as heart failure, pulmonary hypertension, chronic kidney disease, and prior sternotomy—were commonly described qualitatively across studies but were not consistently reported in extractable numeric form, precluding accurate pooled prevalence estimates.

Prior Transvenous Pacing Systems

Across the 34 studies included in this review, LPs were implanted both as de novo devices and in patients with preexisting transvenous pacing systems. Thirteen studies comprising 39 patients reported exclusively de novo LP implantation. In contrast, 18 studies included patients with prior cardiac implantable electronic devices, and ≈67 patients across the pooled cohort had preexisting pacemakers or implanted cardioverter-defibrillator leads. The remaining studies did not consistently specify prior pacing status, reflecting the heterogeneous reporting typical of case reports and small observational cohorts.

Lead-management strategies varied across studies. Transvenous lead extraction before LP implantation was reported in 39 patients across 13 studies, most commonly due to lead-related TR, device infection, lead malfunction, or the need to avoid transvalvular lead passage following tricuspid valve intervention. Abandoned transvenous leads left in situ were reported in 4 patients across 4 studies. In addition, 1 TTVR cohort described transvenous leads intentionally jailed by the prosthetic valve frame (17 patients), although these data were reported at the cohort level, and not all of these patients underwent LP implantation.

Given the heterogeneity of reporting—particularly in case reports and small series—precise pooling of de novo versus prior-lead proportions across the full cohort of 272 patients was not possible. Nevertheless, the available data demonstrate that LP implantation was used both as primary de-novo pacing and as a valve-sparing strategy following lead extraction or in the presence of abandoned or jailed leads.

Tricuspid Valve Interventions

All patients had undergone, or were undergoing, a tricuspid valve intervention at the time of LP implantation.

Surgical interventions included:

- 121 of 272 (44.5%) with prior surgical tricuspid valve repair
- 88 of 272 (32.4%) with a surgical bioprosthetic tricuspid valve
- 3 of 272 (1.1%) with valve-in-valve tricuspid replacement

Transcatheter interventions were also well represented:

- 12 of 272 (4.4%) underwent TEER using TriClip or Pascal systems
- 57 of 272 (21.0%) underwent orthotopic TTVR, predominantly with the EVOQUE system

Across studies, patients frequently exhibited complex tricuspid anatomy, often combining annuloplasty rings, surgical bioprostheses, valve-in-valve constructs, TEER clips, and TTVR frames.

LP Systems

- Micra systems, including Micra ventricular rate-responsive (VR), Micra atrioventricular (AV), Micra transcatheter pacing system (TPS), and Micra V2, were used in 258 of 272 (94.9%) patients.
- Aveir VR systems were used in 5 of 272 (1.8%) patients.
- Nine (3.3%) patients received an unspecified or mixed leadless pacemaker.

In 1 case, an Aveir VR was implanted adjacent to an abandoned Micra VR, illustrating the feasibility of sequential LP strategies in highly complex anatomy.

Access Routes

Access strategy was reported for the majority of patients and reflected both conventional and innovative approaches adapted to challenging venous and valvular anatomy:

- Transfemoral delivery in 142 of 272 (52.2%) patients
- Internal jugular access in 3 of 272 (1.1%) patients, including a case implanted through a Glenn shunt
- Direct surgical right atrial access in 116 of 272 (42.6%) patients, typically during open tricuspid or multivalve surgery

Thus, while transfemoral implantation remained the predominant route, a substantial proportion of LPs were implanted intraoperatively under direct vision, avoiding transvalvular passage of catheters in high-risk surgical populations. Access route was not specified in 11 patients.

Procedural Success and Deployment

The overall procedural success rate was 270 of 272 (99.3%), with 2 of 272 (0.7%) failed implants. Both

Table 2. Characteristics of Included Studies

Study (first author, year)	Year	Country	Design	Sample size
Terricabras, 2021[20]	2021	Canada	Case report	1
Garweg, 2020[21]	2020	Belgium	Retrospective single-center cohort	9
Hale, 2022[22]	2022	USA	Case report	1
Mohamed, 2025[23]	2025	Saudi Arabia	Case report	1
Afzal, 2019[24]	2019	USA	Multicentre retrospective cohort	19
Doyle, 2025[25]	2025	Ireland	Case report	1
Mohammadi, 2025[26]	2025	Iran	Case report	1
Adukauskaite, 2020[27]	2020	Austria	Case report	1
Fink, 2024[28]	2024	Germany	Retrospective single-center case series	12
Morani, 2021[29]	2021	Italy	Case series (2 cases)	2
Ozdemir, 2025[30]	2025	USA	Case report	1
Augustin, 2025[31]	2025	Germany	Case report	1
Rangavajla, 2025[32]	2025	USA	Prospective single-center observational	12
Goto, 2024[33]	2024	Japan	Case report	1
Oates, 2024[34]	2024	USA	Retrospective single-center cohort	99
Patrascu, 2025[35]	2025	Canada	Case report	1
Shivamurthy, 2022[36]	2022	USA	Retrospective single-center cohort	15
Arunthamakun, 2024[37]	2024	USA	Case report (2 cases)	2
Aders and Strobel, 2024[38]	2024	USA	Case report + literature review	1
Demirtakan, 2025[39]	2025	Germany	Case report	1
Vuppuluri, 2025[40]	2025	USA	Case report	1
Poggio, 2024[41]	2024	Italy	Case report	1
Zou, 2025[42]	2025	China	Case report	1
Shen, 2025[43]	2025	China	Case report (2 cases, only 1 with TV bioprosthesis)	1
Dreyfus, 2023[44]	2023	France	Case report	1
Tang, 2017[45]	2017	USA	Case report	1
Braunstein, 2025[46]	2025	USA	Retrospective single-center cohort after TTVR	14
Tat, 2025[47]	2025	USA	Retrospective single-center study (TTVI cohort)	5
Gul, 2024[48]	2024	Multicentre	Retrospective cohort	40
Theis, 2020[49]	2020	Germany	Retrospective cohort	14
Fortuna, 2026[50]	2026	Italy	Case report	1
Friedrich, 2025[51]	2025	Germany	Retrospective cohort	3
Segreti, 2025[52]	2025	Italy	Case report	1
Mekary, 2026[53]	2026	USA	Retrospective cohort	6

Summary of study design, geographic origin, and sample size of all included studies evaluating leadless pacemaker implantation in the context of surgical or transcatheter tricuspid valve interventions. Study designs include case reports, case series, and observational cohort studies. Sample size refers to the number of patients undergoing leadless pacemaker implantation within each study.

TTVI indicates transcatheter tricuspid valve interventions; TTVR, transcatheter tricuspid valve replacement; and TV, tricuspid valve.

failures occurred in patients with extreme TTVR/TEER anatomy in whom the LP delivery system could not be safely maneuvered through the tricuspid apparatus and associated prosthetic structures.

Repositioning or redeployment was variably reported but could be quantified in multiple cohorts and case series:

- In a 12-patient structural series, 1 patient required 3 attempts.
- A 15-patient surgical cohort documented 2 intraoperative redeployments (13.3%).
- Additional case reports described individual redeployments due to suboptimal electrical parameters or deflection against prosthetic frames.

When all extractable data are combined, at least 16 of 272 (5.9%) patients underwent ≥1 redeployment. Because several studies qualitatively described multiple attempts without specifying numbers, this figure likely represents a conservative lower bound.

Acute Electrical Performance

Acute electrical parameters were favorable across the spectrum of tricuspid substrates. Capture thresholds at implantation were typically low, ranging from 0.25 to 1.75 V at standard pulse widths (0.24–0.5 ms), with cohort level means such as:

- 0.36±0.10 V in a 12-patient structural series
- 0.9±1.3 V and 0.9±0.9 V in larger mixed surgical cohorts

A single case reported loss of capture at >5 V in association with late tine fracture and embolization.

R-wave amplitudes at implant ranged from 2 to 18 mV, with most cohort means around 10 to 13 mV, consistent with robust sensing despite previous tricuspid surgery or TTVR. Impedance values spanned 430 to 1490 Ω, with typical cohort means between 600 and 1100 Ω, within the expected range for contemporary LP systems.

Taken together, these data show that acute LP performance is generally preserved even in the presence of annuloplasty rings, bioprostheses, TEER clips, or TTVR frames.

Follow-Up Electrical Parameters

Follow-up duration ranged from early postprocedural interrogation to 12 months of structured clinic or remote monitoring. In the 2 largest cohorts, median follow-up was 12 months (interquartile range, 6–30) and 10.6 months (interquartile range, 2.0–22.7).

Across all cohorts with longitudinal data:

- Capture thresholds remained low and stable, for example, 0.7±0.6 V at 12 months in a 100-patient series, and ≈0.5 V at 6 months in several case reports and small series.
- Sensing amplitudes remained robust, with follow-up R-waves often ≥10 mV in medium-sized cohorts.
- Impedances declined into the expected chronic range (≈500–700 Ω) and remained stable thereafter.

No study demonstrated a pattern of progressive threshold elevation, sensing deterioration, or impedance abnormalities suggestive of chronic LP dysfunction. The only clearly documented late electrical failure was the single case of tine fracture with device embolization.

Interaction With Tricuspid Valve Devices

The interaction between LP systems and tricuspid surgical or transcatheter devices was described in detail in most reports. In nearly all patients, the LP delivery system traversed annuloplasty rings, bioprosthetic valves, valve-in-valve constructs, TEER clips, or TTVR frames without structural damage, provided that fluoroscopic and echocardiographic guidance was used.

Reports highlighted:

- Smooth passage across annuloplasty rings and surgical bioprostheses without leaflet interference or deformation.
- Successful navigation through TEER-created neo-orifices, with deliberate avoidance of clip arms and no clip detachment or leaflet injury.
- Controlled entry through TTVR frames (eg, EVOQUE) with preserved prosthetic function and no frame distortion or obstruction.

Only 2 device-device interaction events were clearly documented:

1. Helix entanglement between an Aveir VR device and a TTVR frame, requiring explantation and reimplantation at a separate site.
2. Failure to implant due to interference from TEER clips, an EVOQUE TTVR frame, and a temporary pacing lead, necessitating alternative pacing.

TR Outcomes

TR behavior largely reflected the underlying tricuspid intervention rather than LP deployment. Across surgical and transcatheter cohorts:

- TEER and TTVR were associated with substantial reductions in TR, for example, from torrential or massive to mild or trace at follow-up.
- In surgical series where TR was reassessed, no deterioration in valve function was attributed to LP implantation.
- In all TEER-LP combinations, no clip detachment or new significant TR was observed after LP placement.

No included study demonstrated new or worsened TR attributable to the LP or its delivery system.

Device-related Complications

Across the entire 272-patient cohort, LP-related complications were rare.

Three LP-specific events were identified (3/272; 1.1%):

- Tine fracture with device embolization to the pulmonary artery, requiring retrieval and reintervention.
- Helix damage and entanglement with a TTVR frame during deployment, necessitating removal and reimplantation.
- Device malfunction requiring LP reimplantation in a cohort patient.

Several pericardial effusions and other postoperative complications were reported in surgical series but were explicitly adjudicated as unrelated to LP implantation. No case of LP-related endocarditis, systemic embolism, or cardiac perforation was reported.

Mortality

Mortality was not numerically reported in all studies. Across available data:

- 13 deaths (13/263; 4.9%) occurred, all attributed to underlying cardiac disease, sepsis, or right ventricular failure rather than LP implantation.
- No death was adjudicated as LP-related in any study.

Because most case reports and small series did not provide systematic survival data, this mortality rate reflects only the subset of patients with reported follow-up outcomes.

DISCUSSION

This systematic review consolidates the largest body of evidence to date evaluating LPs in patients with surgically or percutaneously altered tricuspid valve anatomy. Across 34 studies and 272 patients, the findings consistently demonstrate that LP implantation is safe, technically feasible, and electrically durable—even in the context of prior tricuspid valve repair, bioprosthetic replacement, TEER, valve-in-valve tricuspid replacement, and orthotopic TTVR.[1,6] Importantly, these outcomes address a growing clinical challenge as the prevalence of tricuspid valve disease, structural tricuspid interventions, and postoperative conduction disturbances continues to rise globally.[6,54]

Traditional transvenous pacing is increasingly recognized as a detrimental factor in the progression of TR. Multiple echocardiographic and mechanistic studies demonstrate that transvalvular leads impair leaflet coaptation, promote chordal tethering, induce annular dilation, or become trapped between prosthetic or transcatheter valve components, resulting in clinically significant TR and worsening right-sided heart failure.[3] Meta-analytic evidence supports a strong association between lead-leaflet interference and TR worsening (odds ratio, ≈8.7).[55] Furthermore, right-sided endocarditis and device infections remain a major concern in this population, with transvenous systems contributing to biofilm formation and recurrent bloodstream infections, particularly in patients with prior valve surgery.[4,56] The LP's fully intracardiac configuration eliminates these risks by avoiding both the valvular apparatus and the subcutaneous pocket entirely.[56]

The extremely high implantation success rate in this review (270/272; 99.3%) is consistent with real-world LP data demonstrating procedural success rates exceeding 98% to 99%, even in anatomically challenging cases.[57,58] That this level of success was replicated in patients with annuloplasty rings, bioprosthetic valves, TEER clips, and bulky TTVR frames underscores an important principle: right-sided structural interventions, rather than restricting pacing options, can be safely navigated using modern leadless systems.[4] This advantage is clinically meaningful because TTVR frames, TEER clips, or surgical rings may render transvenous lead placement impossible or unsafe, especially after prosthesis implantation.[4] In many such cases, LPs serve as the only feasible pacing strategy.

An additional observation emerging from this review is the relatively frequent use of intraoperative leadless pacemaker implantation during tricuspid valve surgery. In several surgical cohorts, LPs were implanted under direct visualization by the operating surgeon immediately after valve repair or replacement. This approach avoids the need for epicardial pacing leads, which are traditionally used in this setting but are associated with higher pacing thresholds, reduced sensing reliability, and higher long-term lead failure rates. In contrast, intraoperative LP implantation combines the advantages of direct surgical access with the electrical performance and durability of transvenous pacing systems while preserving the integrity of the repaired or prosthetic tricuspid valve. As experience grows, this strategy may represent a paradigm shift in surgical pacing practice, potentially reducing reliance on epicardial systems in patients undergoing tricuspid valve procedures.

Electrical performance remained excellent both acutely and longitudinally. Capture thresholds were low and stable, sensing amplitudes exceeded guideline-recommended minima, and impedances behaved as expected during chronic follow-up—mirroring performance in general LP populations.[57] These findings indicate that the altered right-sided anatomy following valve intervention does not compromise intracardiac fixation, electrical coupling, or long-term leadless pacing function.[4] Importantly, even in cases requiring multiple deployment attempts, no persistent electrical instability was observed. This is consistent with early Micra performance data and anatomic feasibility work demonstrating that repeated positioning/retrieval and even multiple device accommodation within the right ventricle can be achieved without device-device interaction in tested anatomies.[57,59]

Across the included studies, worsening TR attributable to the leadless pacemaker or its delivery system was not observed. Prior meta-analyses have established that transvenous leads significantly increase the risk of new or worsening TR and associated morbidity.[3,60] In contrast, the valve-sparing design of LPs avoids a transvalvular lead and may reduce the risk of lead-leaflet interaction, although leadless systems are not entirely free from potential tricuspid valve complications.[4] Even in highly complex anatomic substrates—including patients with large coaptation gaps, multiple TEER clips, and malaligned TTVR frames—LPs did not compromise prosthetic or native valve function. This supports emerging consensus recommendations that LPs may represent an attractive valve-sparing pacing strategy in patients with prior tricuspid intervention, especially when transvalvular lead passage may jeopardize repair durability or prosthetic integrity.[4,14] These findings are directionally aligned with contemporary consensus recommendations emphasizing valve-sparing pacing strategies (including leadless pacing) in patients undergoing transcatheter tricuspid interventions.[14] However, because most included studies were case reports or small observational cohorts with relatively short follow-up durations, the long-term impact of leadless pacemaker implantation on tricuspid valve function remains incompletely defined.

However, these findings should be interpreted within the specific context of patients with prior or concomitant tricuspid valve intervention. In the broader leadless pacing literature, results are heterogeneous. Vaidya et al[61] reported less worsening TR with leadless than with transvenous pacing in a real-world cohort, whereas Hai et al[62] showed that close proximity of a septally implanted leadless pacemaker to the tricuspid annulus predicted worse TR, suggesting that device position is mechanistically relevant. More recent comparative data suggest less favorable valvular outcomes with leadless pacing than with left bundle branch area pacing.[63] In addition, rare case reports have described leadless pacemaker-mediated TR, leaflet trauma, leaflet flail, and severe regurgitation requiring surgical intervention.[64–67] Accordingly, our findings support the feasibility and relative valve-sparing advantages of leadless pacing in patients with tricuspid valve intervention, but they do not exclude the possibility of worsening TR related to implantation technique, device position, or patient-specific anatomy.

Complication rates in this review were exceptionally low. Only 3 device-specific events (1.1%) occurred, and all were manageable without long-term sequelae. There were no reports of device-related endocarditis, a clinically significant observation given that the reviewed population included individuals with prior device infection, intravenous drug use, and tricuspid endocarditis—settings where transvenous systems carry substantial infectious risk.[4,56] However, the absence of reported endocarditis or late structural complications should be interpreted cautiously, as most studies were case reports or small cohorts with limited follow-up. No clinically meaningful perforation, embolization, or prosthetic valve injury was described, although these rare events may be underdetected in small observational data sets.[57,58]

Mortality was not numerically reported in all studies. Across available data, 13 deaths occurred, all attributed to underlying cardiac disease, sepsis, or right ventricular failure rather than leadless pacemaker implantation. However, these findings must be interpreted cautiously, as mortality data were derived from studies with short and highly nonuniform follow-up, and systematic long-term survival assessment was not consistently available.[58] Consequently, the reported mortality rates likely underestimate late events and should not be interpreted as definitive evidence regarding long-term survival after leadless pacemaker implantation in this population.[4]

These findings hold important implications for clinical practice. Patients undergoing tricuspid surgery or transcatheter tricuspid interventions frequently develop high-grade atrioventricular block or sick sinus syndrome requiring permanent pacing.[4] Historically, management has been challenging because placement of a transvalvular lead risks damaging a repaired tricuspid valve, interfering with prosthetic valve function, or becoming trapped within TEER clips or TTVR frames.[4,14] In this context, LPs provide a reliable and anatomically compatible pacing strategy that avoids transvalvular hardware. Importantly, the reviewed literature demonstrates that LP implantation is used both as a primary pacing approach following tricuspid valve intervention and as a valve-sparing alternative after transvenous lead extraction or in the presence of abandoned or jailed leads in complex right-sided anatomy.[14] This flexibility makes LP systems particularly attractive in patients with prior tricuspid repair, valve-in-valve procedures, TEER-treated valves, or TTVRs, where transvenous lead placement may compromise procedural durability or future interventions. For example, patients at risk of conduction disturbances after TTVR may benefit from early LP implantation rather than attempting to navigate a conventional pacing lead through a rigid prosthetic frame. Consequently, incorporating leadless pacing into preprocedural planning by the multidisciplinary heart team may improve procedural safety and preserve long-term tricuspid valve function in this growing patient population.[3,4]

From a systems-level perspective, the adoption of LPs in this subset of patients may reduce reoperation rates, prevent valve dysfunction, and decrease right-sided heart failure readmissions—translating to substantial reductions in health care utilization and patient morbidity (supported indirectly by evidence of lower reintervention and chronic complication rates with leadless versus transvenous single-chamber pacing in comparative data sets).[58,68] As structural tricuspid interventions expand, integrating leadless pacing into a unified heart-team strategy may represent a paradigm shift in the management of bradyarrhythmias in complex right-sided anatomy.[4]

Another important consideration is that some patients undergoing tricuspid valve intervention may ultimately require cardiac resynchronization therapy rather than single-chamber ventricular pacing.[69] Conventional cardiac resynchronization therapy systems rely on transvenous right ventricular and coronary sinus leads, which may be technically challenging or undesirable in patients with prior tricuspid repair or replacement. Emerging leadless cardiac resynchronization therapy technologies, such as the WiSE-cardiac resynchronization therapy system, offer the possibility of endocardial left ventricular pacing without transvenous leads and may therefore represent an attractive future strategy for patients with complex right-sided anatomy or prior tricuspid interventions. Although clinical experience with these systems in the setting of tricuspid valve procedures remains limited, further research may help define their role in this population.[70]

LIMITATIONS AND FUTURE DIRECTIONS

This systematic review has several important limitations that should be considered when interpreting its findings. First, the evidence base is predominantly composed of case reports and small observational cohorts, which inherently limits the strength of causal inference. Case reports—while clinically valuable for characterizing device-valve interactions in highly complex anatomies—tend to overrepresent successful procedures and underreport neutral or unsuccessful experiences.[71,72] The absence of controlled comparisons means that the true relative safety and efficacy of LPs versus alternative pacing strategies (epicardial, coronary sinus, or His/left bundle pacing) cannot be definitively established.

Second, substantial heterogeneity exists in reporting quality across studies. Key clinical variables—including baseline TR severity, right ventricular function, pulmonary pressures, and pacemaker dependency—were often incompletely described. Similarly, electrical performance was not uniformly reported across acute, short-term, and long-term intervals. This heterogeneity limits the ability to pool data, increases the risk of reporting bias, and mirrors broader deficiencies in device-related reporting for structural right-sided interventions.[73,74] Existing electrophysiology and device-specific reporting standards emphasize the need for comprehensive baseline characterization, procedural detail, and longitudinal outcome reporting to enable meaningful comparison across studies, yet adherence to these frameworks was inconsistent across the included literature.[75]

Third, follow-up duration varied widely, and long-term (>2 years) performance in patients with surgical rings, bioprosthetic valves, TEER devices, and TTVR prostheses remains insufficiently understood. Although LPs have demonstrated excellent long-term electrical stability and low complication rates in general populations, with follow-up extending to 5 years and beyond,[76,77] the unique mechanical and hemodynamic environment created by right-sided prostheses may confer unrecognized late risks.[78] The long-term implications of LP extraction, device battery depletion, and end-of-service management in patients with tricuspid prostheses also remain unknown—particularly because transcatheter prostheses may impede future device retrieval.[79] While chronic retrieval of helix-fixation LPs has shown high success rates (≈90%) even after prolonged implant durations, retrieval feasibility through TTVR frames or in the presence of TEER devices has not been systematically evaluated.[80] In particular, the long-term interaction between LPs and repaired or prosthetic tricuspid valves has not been systematically studied, and delayed effects on valve function or right-sided hemodynamics cannot be excluded.

Fourth, publication bias is likely. Successful LP implantation in challenging posttricuspid anatomies is far more likely to be reported than unsuccessful or aborted procedures. This imbalance may underestimate procedural risks and device-prosthesis interaction complications. Such selective reporting has been widely documented in structural heart intervention literature, including TEER, TTVR, and left-sided transcatheter valve studies, where early experience often preferentially highlights favorable outcomes and underrepresents procedural failure or late complications.[81]

Fifth, although this review includes the largest number of LPs implanted in the context of tricuspid intervention to date, the total sample size (n=272) remains modest relative to the global population undergoing tricuspid repair, replacement, or TTVR. The absence of randomized trials or large multicenter prospective cohorts limits the generalizability of the findings. Future studies adopting standardized end point definitions—such as those proposed by the Tricuspid Valve Academic Research Consortium—would substantially improve cross-study comparability and robustness of safety and efficacy assessment.[6]

Future research should focus on addressing these knowledge gaps. Prospective multicenter registries and randomized comparisons are needed to evaluate LPs against epicardial and coronary sinus pacing strategies in patients undergoing tricuspid intervention. Dedicated trials in TTVR populations are particularly important, given the high incidence of postprocedural conduction disturbances, with new-onset high-grade atrioventricular block reported in 13.5% to 44% of patients and permanent pacemaker implantation required in ≈13% to 25% of cases following contemporary TTVR systems. Advanced imaging studies using 3-dimensional transesophageal echocardiography, intracardiac echocardiography, and cardiac computed tomography should systematically characterize device-valve interaction during implantation and follow-up. Mechanistic studies evaluating the impact of LP positioning (septal versus apical versus inflow) on right ventricular remodeling, prosthesis function, and TR progression are also needed.[82,83]

Long-term investigations examining management at battery end-of-life—including feasibility, safety, and strategies for second-device implantation—are essential to establish comprehensive guidance for lifelong pacing in this population.[84] With median projected battery longevity ranging from 14 to 17 years in real-world LP cohorts and successful chronic retrieval reported up to 9 years postimplantation, optimal end-of-service strategies (retrieval versus abandonment with new implantation) remain undefined, particularly in patients with tricuspid prostheses where retrieval may be technically constrained.[80]

Finally, future studies should explore the cost-effectiveness of LPs in this population. While LPs have demonstrated favorable cost-effectiveness compared with transvenous systems in general pacing populations, with incremental cost-effectiveness ratios in the range of €5000 to €6500 per quality-adjusted life year, the economic implications in patients with tricuspid valve interventions—who face higher risks of lead-related complications and reintervention—remain inadequately studied.[85]

Collectively, these limitations highlight the need for structured, longitudinal, and high-quality evidence to guide optimal pacing management in patients with surgically or transcatheter-altered tricuspid valve anatomy. Until such data are available, the favorable safety and performance signals demonstrated in this review support the continued use of LPs as an important valve-sparing pacing option in this anatomically and clinically complex population.

CONCLUSIONS

In this systematic review—the most comprehensive synthesis to date of leadless pacemaker implantation in patients with surgically or transcatheter-altered tricuspid valve anatomy—leadless pacing showed consistently high procedural success, robust electrical performance, and very low device-related complication rates across a wide range of complex structural scenarios. Among the included studies, no worsening TR was attributed to the device, and interactions with annuloplasty rings, bioprosthetic valves, TEER repairs, and TTVRs were generally safe when implantation strategies were tailored to the underlying anatomy. Nevertheless, reports from the broader leadless pacing literature indicate that tricuspid valve injury or worsening regurgitation may rarely occur in selected cases.

These findings highlight LPs as a reliable, valve-sparing solution for permanent pacing in a population in which conventional transvenous leads carry substantial risk. As tricuspid valve interventions expand and lead-associated TR becomes increasingly recognized, leadless pacing offers an important valve-sparing option in the management of bradyarrhythmias in this high-risk cohort. Future prospective studies, standardized reporting frameworks, and long-term follow-up are needed to refine patient selection, optimize implantation technique, and clarify the role of leadless systems in patients undergoing or having undergone tricuspid valve repair or replacement.

ARTICLE INFORMATION

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